← Back **History: Dawn of Electronic Computing**
Graded Assignment • 30 min

Due Mar 9, 11:59 PM CET

1. What was the Lorenz Machine used to transmit? 1 point

- ○ Long strategic messages with lots of detail
- ○ Attack plans between the British and Russian generals
- ○ Intelligence between the American and British forces
- ◉ News stories from Germany to the soldiers at the front so they would feel more connected to family and friends back home

2. Who is John Forbes Nash and what is he known for? 1 point

- ○ An actor in the movie A Beautiful Mind
- ○ The creator of the internet
- ◉ A mathematician and founder of modern day Game Theory
- ○ A famous spy during World War II

3. What did the Polish Cipher Bureau give to the British? 1 point

- ○ A German Geheimschreiber
- ○ A Polish dessert called the Bomba
- ○ Access to the Polish telephone network data transmissions
- ◉ A technique for breaking encoded German Enigma messages

4. Why was the Enigma machine so important to the German strategy during World-War II? 1 point

- ○ Because it allows data to be stored on punched cards and quickly read so that computer programs could easily be changed.
- ○ Because Germany's enemies would be distracted if they believed all the mis-information in the communication
- ○ World War II was widely distributed and could not depend on wired telephone and telegraph for communications
- ○ Because it was the quickest way to insure that as many people as possible would be exposed to propaganda messages.

5. What is a Modem used for? 1 point

- ○ Use a voice-based phone line to transmit data
- ○ Decrypt coded German war time transmissions
- ○ Insure that transmissions to submarines work even when they are deep under water
- ○ Record encrypted data from wireless transmissions for later decryption.
- ○ Retransmit lost packets so as to insure the overall reliability of Internet connections

6. What kind of parts were used to make the Colossus electronic computer? 1 point

- ○ Vacuum tubes, relays, switches, and lights
- ○ Microprocessors and random access memory
- ○ A gas turbine that drove an air compressor
- ○ A complex series of gears, pulleys and springs that allowed it to walk forward in a halting motion

7. What are leased lines? 1 point

- ○ The modern lines we connect to the internet through
- ○ Dedicated telephone lines organizations paid telecom companies monthly to have continuous access to
- ○ Secure lines used to connect British cryptographers to British military command

8. About how many vacuum valves/tubes were there in the Colossus? 1 point

- ○ 10000
- ○ 2500
- ○ 150
- ○ 12

9. On the Colossus computer what was used to store and repeatedly read the encrypted message text? 1 point

- ○ Strips of film
- ○ A spinning magnetic disk drive that was read like a tape
- ○ A solid-state flash drive (i.e. a USB stick)
- ○ A paper tape that was read using light sensitive tubes

10. What made Bletchley park successful? 1 point

- ○ The information provided by the Polish Cipher Bureau
- ○ Their huge team dedicated to the purpose of decryption
- ○ The unlimited budget available to them
- ○ All of the above

11. Who did the Enigma machine belong to? 1 point

- ○ The Polish
- ○ The British
- ○ The Norwegians
- ○ The Germans

12. Who created the functional design of the Bombe mechanical computer use to crack Enigma codes at Bletchley Park? 1 point

- ○ Alistair Dennison and Tommy Flowers
- ○ Max Newman and William (Bill) Tutte
- ○ Alan Turing and Gordon Welchman

13. Where is Bletchley Park located? 1 point

- ○ Glasgow, Scotland
- ○ Buckinghamshire, England
- ○ Sydney, Australia
- ○ New Mexico, United States